

# DaTra: Data Transformation Semantics as a Symmetric Monoidal Category on Atlas Presheaves

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## Abstract

This work formalizes the semantics of the Datra programming-language project in Lean. Data domains are organized into atlases: structured, eventually stable families of pages encoding how data is partitioned and related. Data transformations are modeled as presheaves on the resulting atlas categories, yielding a topos-theoretic semantics. Stable transformations are then assembled over atlas federations, whose horizontal sum extends by Day convolution to a symmetric monoidal category.

## 1 Dominions

**Definition 1.1** (The Category of Dominions). The **Category of Dominions**, denoted  $\mathbf{Dom}$ , has as its objects sets whose cardinality is at most  $\omega_0$ , equivalently sets admitting an injection into  $\omega_0$ , and as its morphisms set-theoretic functions.

**Definition 1.2** (The Domanial Embedding Functor). The **Domanial Embedding Functor**, denoted  $\mathbf{Emb} : \mathbf{Dom} \rightarrow \mathbf{Set}$ , sends each dominion to its underlying set.

## 2 Domanial Insertions

**Definition 2.1** (The Category of Domanial Insertions). The **Category of Domanial Insertions**, denoted  $\mathbf{DomIns}$ , has dominions as objects and monomorphisms in  $\mathbf{Dom}$  as morphisms.

**Definition 2.2** (The Category of Codomanial Insertions). The **Category of Codomanial Insertions**, denoted  $\mathbf{CoDomIns}$ , is the opposite category  $\mathbf{DomIns}^{\text{op}}$ .

## 3 Chains

**Definition 3.1** (Chain). A **Chain** is a totally ordered thin category, indexed skeletally by an ordinal strictly smaller than  $\omega_0^{\omega_0}$ .

**Definition 3.2** (The Spine). The **spine** is the chain with  $\omega_0$  objects.

## 4 Consolidations

**Definition 4.1** (The Category of Consolidations). The **Category of Consolidations**, denoted  $\mathbf{Con}$ , has chains as objects and point-surjective functors between their thin categories as morphisms.

**Definition 4.2** (The Category of Coconsolidations). The **Category of Coconsolidations**, denoted  $\mathbf{CoCon}$ , is the opposite category  $\mathbf{Con}^{\text{op}}$ .

**Definition 4.3** (Consolidation Transport). The **Consolidation Transport Functor** is denoted  $\mathbf{ConTra}$ . It is a functor from  $\mathbf{Con}$  to  $\mathbf{Set}$  that sends a chain to its object set and a consolidation to its underlying set-theoretic surjection.

## 5 Folios and Paginations

**Definition 5.1** (Folio). A **folio** is a functor  $F : S \rightarrow \mathbf{CoCon}$ , where  $S$  is the spine, such that  $F(0)$  is the singleton chain.

**Definition 5.2** (Cell Occurrences). For a folio  $F$ , its **category of cell occurrences**, denoted  $\mathbf{El}(F)$ , has as objects pairs  $(m, k)$ , where  $m$  is a page and  $k$  is an object of the chain  $F(m)$ . An arrow  $(m, k) \rightarrow (n, l)$  is an arrow  $f : m \rightarrow n$  in the opposite spine for which the consolidation induced by  $F$  transports  $k$  exactly to  $l$ . Equivalently,

$$\mathbf{El}(F) = \int (\mathbf{ConTra} \circ F^{\text{op}}).$$

**Definition 5.3** (The Category of Paginations). The **Category of Paginations**, denoted  $\mathbf{Pag}$ , has as objects a folio  $Fo$  together with its associated category of cell occurrences  $El = \mathbf{El}(Fo)$ . For  $W : \mathbf{Pag}$ , write  $W_{Fo}$  for its folio and  $W_{El}$  for its occurrence category. A morphism  $T : X \rightarrow Y$  in  $\mathbf{Pag}$  is a functor  $T : X_{El} \rightarrow Y_{El}$ .

## 6 Atlases

**Definition 6.1** (The Category of Atlases). The **Category of Atlases**, denoted  $\mathbf{Atl}$ , has as objects pairs  $(Pa, Da)$ , where  $Pa : \mathbf{Pag}$  and  $Da : Pa_{El} \rightarrow \mathbf{Domlns}$  is a functor, equivalently a presheaf  $Da : Pa_{El}^{\text{op}} \rightarrow \mathbf{CoDomlns}$ . For the folio  $Pa_{Fo}$ , a page  $m$ , and distinct objects  $k, k'$  of the chain  $Pa_{Fo}(m)$ , the pullback in  $Da$  of the arrows induced by  $(m \rightarrow 0)(k)$  and  $(m \rightarrow 0)(k')$  is empty.

For  $X : \mathbf{Atl}$ , write  $X_{Pa}$  and  $X_{Da}$  for its two components, and abbreviate  $X_{Fo} := (X_{Pa})_{Fo}$  and  $X_{El} := (X_{Pa})_{El}$ .

For every  $n \in \mathbb{N}$ , the  $n$ **th atlas page** of  $X$  is the chain  $X_{Fo}(n)$  on its full spine. An object  $k$  of this chain is a **page cell**, written  $(n, k)$ , and the dominion carried by that cell is  $X_{Da}(n, k)$ .

There is an integer  $w$  such that, for every  $w' > w$ , the consolidation from page  $w'$  to page  $w$  induced by  $Pa_{Fo}$  is the identity and, for every object  $k$  of  $Pa_{Fo}(w)$ , the corresponding arrow  $Da((w' \rightarrow w)(k))$  is the identity. The least such integer is the **cardinality of the atlas**, denoted  $|A|$ .

A morphism  $T : X \rightarrow Y$  is a pair  $(T_{Pa}, T_{Da})$ , where  $T_{Pa} : X_{El} \rightarrow Y_{El}$  is a functor and  $T_{Da} : X_{Da} \Rightarrow Y_{Da} \circ T_{Pa}$  is a natural transformation. Thus, for every  $f : x \rightarrow y$  in  $X_{El}$ , the following square commutes:

$$\begin{array}{ccc} X_{Da}(x) & \xrightarrow{X_{Da}(f)} & X_{Da}(y) \\ (T_{Da})_x \downarrow & & \downarrow (T_{Da})_y \\ Y_{Da}(T_{Pa}(x)) & \xrightarrow{Y_{Da}(T_{Pa}(f))} & Y_{Da}(T_{Pa}(y)). \end{array}$$

Finally, if  $x = |X|$ ,  $x' > x$ , and  $r$  is an object of the final chain  $X_{Fo}(|X| - 1)$ , then  $T_{Pa}(x', r) = T_{Pa}(x, r)$  and  $(T_{Da})_{(x', r)} = (T_{Da})_{(x, r)}$ .

**Definition 6.2** (Extent of an Atlas). The **extent** of  $A$  is  $\mathbf{Ex}(A) = A_{Da}(0, 0)$ . Since objects of  $\mathbf{Domlns}$  are dominions,  $\mathbf{Ex}(A) : \mathbf{Dom}$  for every  $A : \mathbf{Atl}$ .

**Definition 6.3** (Territory of an Atlas). Let  $M$  be the objects of the final chain  $A_{Fo}(|A| - 1)$ . The **territory** is the indexed family  $\mathbf{Ter}(A) : M \rightarrow \mathbf{Dom}$  given by  $\mathbf{Ter}(A)(k) = A_{Da}(|A| - 1, k)$ .

**Definition 6.4** ( $n$ th Region of an Atlas). The  $n$ **th region** of  $A$  is  $\mathbf{Ter}(A)(n)$ , with indexing beginning at 0.

## 7 Transposals and Traversals

**Definition 7.1** (The Category of Atlas Transposals). The **Category of Atlas Transposals**, denoted  $\text{AtlTrap}$ , is the wide subcategory of  $\text{Atl}$  whose morphisms  $F : X \rightarrow Y$  have point-injective object maps  $F_{Pa}$ .

**Definition 7.2** (The Category of Atlas Traversals). The **Category of Atlas Traversals**, denoted  $\text{AtlTrav}$ , is the wide subcategory of  $\text{AtlTrap}$  whose morphisms satisfy the following conditions. For  $F : X \rightarrow Y$ , let  $x = (m, i)$  and  $y = (m', j)$ , put  $p = \min(m, m')$ , write  $F_{Pa}(x) = (n, i')$  and  $F_{Pa}(y) = (n', j')$ , and put  $p' = \min(n, n')$ . Let  $c_{m,p}^X$  and  $c_{m',p}^X$  denote the consolidations induced by the folio  $X_{Fo}$ , and define the analogous maps for  $Y$ . If

$$c_{m,p}^X(i) < c_{m',p}^X(j),$$

then

$$c_{n,p'}^Y(i') < c_{n',p'}^Y(j').$$

Furthermore, if  $q \in |X|$  and  $t \in X_{Da}(q)$  is in the image of a final region—that is, there exist  $k \in |\text{Ter}(X)|$  and  $l \in \text{Ter}(X)(k)$  whose image under  $X_{Da}((|X| - 1) \rightarrow q)(k)$  is  $t$ —then this property is preserved by  $F$ .

**Definition 7.3** (The Category of Stable Atlas Traversals). The **Category of Stable Atlas Traversals**, denoted  $\text{StaAtlTrav}$ , is the wide subcategory of  $\text{AtlTrav}$  whose morphisms  $F : X \rightarrow Y$  satisfy  $F_{Pa}(0, 0) = (0, 0)$ .

## 8 Maps and Cartography

**Definition 8.1** (The Category of Atlas Maps). The **Category of Atlas Maps**, denoted  $\text{AtlMap}$ , is the full subcategory of atlases for which every element of the extent is in the image of a final region.

**Definition 8.2** (The Atlas Map Inclusion Functor). The **Atlas Map Inclusion Functor**  $\text{AtlMapInc} : \text{AtlMap} \rightarrow \text{Atl}$  is the canonical inclusion.

**Definition 8.3** (The Category of Atlas Traversal Maps). The **Category of Atlas Traversal Maps**, denoted  $\text{AtlTravMap}$ , is the full subcategory of  $\text{AtlTrav}$  whose objects are Atlas Maps.

**Definition 8.4** (The Atlas Traversal Map Inclusion Functor). The canonical inclusion is denoted  $\text{AtlTravMapInc} : \text{AtlTravMap} \rightarrow \text{AtlTrav}$ .

**Lemma 8.5** (Cartography Lemma). *The inclusion  $\text{AtlTravMapInc}$  has a right adjoint, denoted the **Charting Functor**  $\text{Chr} : \text{AtlTrav} \rightarrow \text{AtlTravMap}$ .*

*Proof sketch. On an object  $X : \text{AtlTrav}$ , the functor  $\text{Chr}$  sends  $X$  to the universal arrow from  $\text{AtlTravMapInc}$  to  $X$ . Write  $X' = \text{AtlTravMapInc}(\text{Chr}(X))$ , together with  $H : X' \rightarrow X$ , with  $X'$  terminal among objects having this property. The solution is the  $\text{DaTra}$  map satisfying  $\text{Ter}(X') \cong \text{Ter}(X)$ . It is unique because  $\text{AtlTrav}$  preserves orders and terminal because  $H_{Pa}$  is faithful.*

## 9 Coalitions

**Definition 9.1** (The Coalizing Functor). The **Coalizing Functor**, denoted

$$\text{Coa} : \text{StaAtlTrav} \longrightarrow \text{DomIns},$$

is the extent of the charted atlas:  $\text{Coa} = \text{Ex} \circ \text{Chr}$ .

**Definition 9.2** (The Coalition of an Atlas). The **coalition** of an atlas  $X$  is the extent of its chart, regarding  $X$  as an object of the wide category of stable traversals.

**Definition 9.3** (The Domanial Inclusion Functor). The **Domanial Inclusion Functor** sends a dominion to the atlas of cardinality 1 whose coalition is that dominion. Write  $\text{DomInc} : \text{DomIns} \rightarrow \text{StaAtlTrav}$  for this functor. It is left adjoint to  $\text{Coa}$ : for  $X : \text{DomIns}$  and  $Y : \text{StaAtlTrav}$ , naturally in  $X$  and  $Y$ ,

$$\text{Hom}_{\text{StaAtlTrav}}(\text{DomInc}(X), Y) \cong \text{Hom}_{\text{DomIns}}(X, \text{Coa}(Y)),$$

and  $\text{Coa}(\text{DomInc}(X)) \cong X$ .

## 10 Data Transformations

**Definition 10.1** (The Category of Data Transformations). The **Category of Data Transformations**, also called the category of **Data Transformation Sets** or **DaTra Sets**, is the presheaf category

$$\text{DaTra} = [\text{Atl}^{\text{op}}, \text{Set}].$$

As a presheaf category, it is a topos.

**Definition 10.2** (Navigation). A **navigation** of  $D : \text{DaTra}$  is a monomorphism  $\text{Nav} : \text{Yo}(A) \rightarrow D$  for some atlas  $A$ .

**Definition 10.3** (Expedition). An **expedition** is a navigation represented by an Atlas Map.

**Definition 10.4** (The Category of Data Transposals). The **Category of Data Transposals**, denoted  $\text{DaTrap}$ , is the presheaf category  $[\text{AtlTrap}^{\text{op}}, \text{Set}]$ .

**Definition 10.5** (The Category of Data Traversals). The **Category of Data Traversals**, denoted  $\text{DaTrav}$ , is the presheaf category  $[\text{AtlTrav}^{\text{op}}, \text{Set}]$ .

**Definition 10.6** (Data Transformation Maps). The **Category of Data Transformation Maps**, denoted  $\text{DaTraMap}$ , is the full subcategory of  $\text{DaTra}$  Sets all of whose navigations are expeditions.

## 11 Atlas Federations

**Definition 11.1** (Atlas Merge). The **Atlas Merge** is the object operation

$$\text{AtlMerge} : \text{Ob}(\text{Atl}) \times \text{Ob}(\text{Atl}) \longrightarrow \text{Ob}(\text{Atl}).$$

For atlases  $X$  and  $Y$ , the extent of  $\text{AtlMerge}(X, Y)$  is the disjoint union of their extents. Both atlases are evaluated on their full spines: page  $n + 1$  of the merge is the tagged, componentwise combination of page  $n$  of  $X$  and page  $n$  of  $Y$ .

**Definition 11.2** (Atlas Federation). An **Atlas Federation** is a countable tagged family of atlas objects. The category of Atlas Federations is denoted  $\text{AtlFed}$ . A morphism consists of a map of tags and, at each source tag, a stable atlas traversal to its selected target tag.

**Definition 11.3** (The Empty Atlas). The **Empty Atlas** is  $\text{Atl} = \text{DomInc}(\emptyset)$ , also viewed as an Atlas Federation.

**Definition 11.4** (The Atlas Horizontal Sum Bifunctor). The **Atlas Horizontal Sum Bifunctor**, denoted  $\text{AtlHorSum} : \text{AtlFed} \times \text{AtlFed} \rightarrow \text{AtlFed}$ , has object action

$$\text{AtlHorSum}(X, X') = \text{AtlMerge}(X, X').$$

Here  $\text{AtlFed}$  is the category of Atlas Federations; its empty federation represents  $\text{Atll}$ . Given stable traversals  $F : X \rightarrow Y$  and  $F' : X' \rightarrow Y'$ , horizontal sum preserves the left and right tags and applies  $F$  and  $F'$  componentwise. Stability fixes the common origin, so this arrow action is again a stable atlas traversal.

**Lemma 11.5** (Horizontal Lemma). *The Atlas Horizontal Sum Bifunctor  $\text{AtlHorSum}$  exists.*

*Proof.* Identity arrows act identically on every tag, composition holds componentwise, and every component arrow is required to lie in  $\text{StaAtlTrav}$ . The federation structure supplies the associator, unitors, and braider by reassociation, deletion of the empty tag family, and tag swapping, respectively.

**Definition 11.6** (The Atlas Braider). The **Atlas Braider**  $\text{AtlBrd}_{F,F'} : \text{AtlHorSum}(F, F') \rightarrow \text{AtlHorSum}(F', F)$  swaps the left and right component tags. Hence  $\text{AtlBrd}_{F,F'} \circ \text{AtlBrd}_{F',F} = \text{Id}$ .

**Definition 11.7** (The Atlas Associator). The **Atlas Associator**, denoted

$$\text{AtlAsoc}_{F,F',F''} : \text{AtlHorSum}(\text{AtlHorSum}(F, F'), F'') \longrightarrow \text{AtlHorSum}(F, \text{AtlHorSum}(F', F'')),$$

flattens the nested federation and retags its three sides using the canonical reassociation  $((x + y) + z) \leftrightarrow (x + (y + z))$ . Its data component is the matching reassociation of the three extent summands.

**Definition 11.8** (The Atlas Unitors). The **Atlas Left Unitor**  $\text{Lu} : \text{AtlHorSum}(\text{Atll}, F) \rightarrow F$  and the **Atlas Right Unitor**  $\text{Ru} : \text{AtlHorSum}(F, \text{Atll}) \rightarrow F$  remove the empty tag family.

## 12 Stable Data Transversals

**Definition 12.1** (Stable Data Transversals). The **Category of Stable Data Transversals**, denoted  $\text{StaDaTrav}$ , is the presheaf category on Atlas Federations whose component arrows lie in  $\text{StaAtlTrav}$ .

**Definition 12.2** (The Empty Map). The **Empty Map**, denoted  $I$ , is the Day unit represented by the empty Atlas Federation:

$$I = \text{Yo}(\text{Atll}).$$

**Definition 12.3** (The Horizontal Sum Bifunctor). The **Horizontal Sum Bifunctor**, denoted  $\text{HorSum} : \text{StaDaTrav} \times \text{StaDaTrav} \rightarrow \text{StaDaTrav}$ , or in infix notation by  $\boxplus : \text{StaDaTrav} \times \text{StaDaTrav} \rightarrow \text{StaDaTrav}$ , is the Day convolution extension of stable atlas horizontal sum. Because the indexing morphisms are arrows of  $\text{StaAtlTrav}$ , its result and its arrow action land in  $\text{StaDaTrav}$  by construction.

**Definition 12.4** (The Stable Data Transversal Braider). The **Stable Data Transversal Braider**  $\text{Brd}_{F,F'} : F \boxplus F' \rightarrow F' \boxplus F$  is the Day convolution extension of  $\text{AtlBrd}$ .

**Definition 12.5** (The Stable Data Transversal Associator). The **Stable Data Transversal Associator**

$$\text{Asoc}_{F,F',F''} : (F \boxplus F') \boxplus F'' \longrightarrow F \boxplus (F' \boxplus F'')$$

is the Day convolution extension of  $\text{AtlAsoc}$ .

**Definition 12.6** (The Stable Data Transversal Unitors). The **Stable Data Transversal Left Unitor**  $\text{Lu} : I \boxplus F \rightarrow F$  and the **Stable Data Transversal Right Unitor**  $\text{Ru} : F \boxplus I \rightarrow F$  are the Day convolution extensions of  $\text{AtlLu}$  and  $\text{AtlRu}$ .

**Definition 12.7** (Stable Data Transversals Monoidal Category). The **Stable Data Transversals Monoidal Category**, denoted  $\text{StaDaTravMon}$ , is

$$\text{StaDaTravMon} = (\text{StaDaTrav}, \boxplus, I).$$

Its tensor is Day convolution on the Atlas Federation. The braider, associator, and unitors are induced by retagging that form, so all coherence maps remain within stable data transversals.